like it. “One of the reasons why people are saying, ‘Oh we should go for the hundred-TeV collider’ is that then you actually get a chance at nailing this thing down.”

As David points out, we already have a puzzle on the table: the nature of the Higgs field, and its (and our) fate. The data we’ve already obtained, and are working to analyze, could begin to trace out the nature of the Higgs in more detail, but with a new collider, we might finally answer the question of what the instability that threatens us with vacuum decay really means.

As we discussed in Chapter 6, the Higgs potential is the mathematical structure that determines how the Higgs field evolves, and, importantly for us, whether it will send us all to our doom. It is, in a real sense, the holy grail of particle physics. But with current theories we have very little handle on what it looks like. Based on our current understanding, its shape depends sensitively on the competing influences of several different hard-to-calculate aspects of the Standard Model, and if some higher-energy theory exists, this could completely change the picture.

Some researchers I spoke to, including CERN theorist (and leading supersymmetry advocate), John Ellis, suspect that the apparent instability of the Higgs is not really an existential threat, but rather a sign that there’s something about the theory that we don’t understand.

José Ramón Espinosa, a theorist studying vacuum decay, hopes to find ways to better understand the Higgs potential and what our precarious placement on the knife edge of stability might mean, without simply waiting for a true vacuum bubble to show up.[fn6] “There’s no reason for the potential to be like this,” he says. “We live in this very, very special place. So for me this is kind of intriguing; maybe this is trying to tell us something.” The key to understanding the Higgs potential ultimately depends on what are called *running couplings*—the interactions between particles and fields and how they change with higher-energy collisions. “This might be one of the main messages of the LHC if we don’t find anything else,” Espinosa says. “Of course, if the LHC finds new physics, then most likely this is going to interfere with the running of the couplings. Then anything can happen. Maybe the potential is stable, maybe it’s even more unstable. We don’t know.”

In addition to the small (but important!) point of determining the fate of the cosmos, a better understanding of the Higgs field could show us how mass works, or why the fundamental forces manifest with the strengths we